Semantic Resume Matcher

AI-Powered Candidate Ranking Pipeline

Core Workflow Overview

1. **Document Ingestion**: Upload Job Descriptions (JD) and Candidate Resumes (PDF).
2. **Text Parsing**: PDF text is extracted via OCR/PDF parsers.
3. **Information Extraction**: LLM processes unstructured text into structured JSON.
4. **Semantic Matching**: High-dimensional embeddings evaluate similarity.
5. **Reranking & Filtering**: Pipeline retrieves top matches and filters out unqualified candidates.

Information Extraction Model

Model Used: `gemini-3.8-flash` (Google GenAI)

- **Job Description Extraction**:
  - Automatically identifies: Required Skills, Preferred Skills, Minimum Experience, Education, and Location Policy.
- **Resume Extraction**:
  - Synthesizes candidate history into discrete skills, years of experience, and educational background.
- **Resilience**: Features an offline fallback mechanism using SpaCy/Regex if API limits are reached.

Semantic Matching Pipeline

Model Used: `sentence-transformers/all-MiniLM-L6-v2` (HuggingFace)
Vector Store: FAISS (Facebook AI Similarity Search)

How it works:

1. **Embedding**: Candidate skills are converted into dense vector representations.
2. **Indexing**: FAISS creates an optimized Cosine similarity index.
3. **Retrieval**: For each JD requirement, the system fetches the nearest neighbor candidate profiles in milliseconds.

Fairness Audit & Explainability

- **Explainable AI**: Generates a natural language explanation detailing *why* a candidate matched (e.g., "Candidate has 5 years of Python, matching JD requirement of 3 years").
- **Fairness Audit**: Evaluates the shortlist to ensure there is no inherent bias based on non-critical demographic or stylistic factors.
- **Score Breakdown**: Provides individual scores for Requirements, Experience, and Skills.

Tech Stack

- **Frontend**: Streamlit with custom Dark Glassmorphism CSS.
- **Backend Logic**: Python 3
- **Generative AI**: Google Gemini 3.8 Flash
- **Embeddings**: HuggingFace SentenceTransformers
- **Vector DB**: FAISS (Facebook AI Similarity Search)
- **Deployment**: Local / Cloud ready